Sl. No. 10211

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JSS Mahavidyapeetha

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JSS SCIENCE AND TECHNOLOGY UNIVERSITY MYSURU

BLUE BOOK

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Roll No. 55 Sem III Sec 'B'

Course Name Computer Organization

Course Code 20CS340

USN No. 01JST20CS160

CIE Event No.	Maximum Marks				
	CO1	CO2	CO3	CO4	CO5
1					
2					
3					
4					
5					
Total					

Course Outcome Statements	
CO1	
CO2	
CO3	
CO4	
CO5	

CIE Event No.	Date	Marks Scored					Final CIE	
		CO1	CO2	CO3	CO4	CO5	Max.	Score
1								
2								
3								
4								
5								
Total								
CO Attainment								

SriramaBhat

Sign. of the Student

Name :

Sign. of the Faculty :

Name :

Sign. of the HOD :

Computer Organization Assignment

1.) Represent the decimal values +14, -19 and -23, as, signed 7-bit numbers in following binary formats
(i) Sign and magnitude (ii) 1's complement (iii) 2's complement

Ans: 7-bit binary value of

14 = 2|14 = 0001110
2|7-0
2|3-1
1-1

19 = 2|19 = 0010011
2|9-1
2|4-1
2|2-0
1-0

23 = 2|23 = 0010111
2|11-1
2|5-1
2|2-1
1-0

Format \ Number	+14	-19	-23
(i) Sign and Magnitude	0001110	1010011	1010111
(ii) 1's complement	0001110	1101100	1101000
(iii) 2's complement	0001110	1101101	1101001

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2.) Write RISC style program to read a line of text from standard input device and display the same on standard output device using polling scheme.

Ans: Let the keyboard be the standard input device and the display be the standard output device.

Program:

```
ILOC:   Subtract    SP, SP, #8
        Store       R2, 4(SP)
        Store       R3, (SP)
        Load        R2, PNTR
        LoadByte    R3, KBD_DATA
        StoreByte   R3, (R2)
        Add         R2, R2, #1
        Store       R2, PNTR

ECHO:   LoadByte    R2, DISP_STATUS
        And         R2, R2, #4
        Branch_if_[R2]=0  ECHO
        StoreByte   R3, DISP_DATA
        Move        R2, #CR
        Branch_if_[R3]≠R2  RTNR
        Move        R2, #LF
        Store       R2, EOL
        Clear       R2
        Store Byte  R2, KBD_CONT
```

```
RNTR:   Load            R3, (SP)
        Load            R2, 4(SP)
        Add             SP, SP, #8
        Return_from_interupt
```

Main Program:

```
START:  Move            R2, #LINE
        Store           R2, PNTR
        Clear           R2
        Store           R2, EOL
        Move            R2, #2
        StoreByte       R2, KBD_CONT
        MoveControl     R2, IENABLE
        Or              R2, R2, #2
        Move Control    IENABLE, R2
        MoveControl     R2, PS
        Or              R2, R2, #1
        Move Control    PS, R2
```

3.) Write an assembly routine to unpack a packed BCD digit stored in PACKED location and store the ASCII values of the unpacked BCD digits in LOC and LOC+1 using shift and logic instructions (Eg: 68 to 36 and 38)

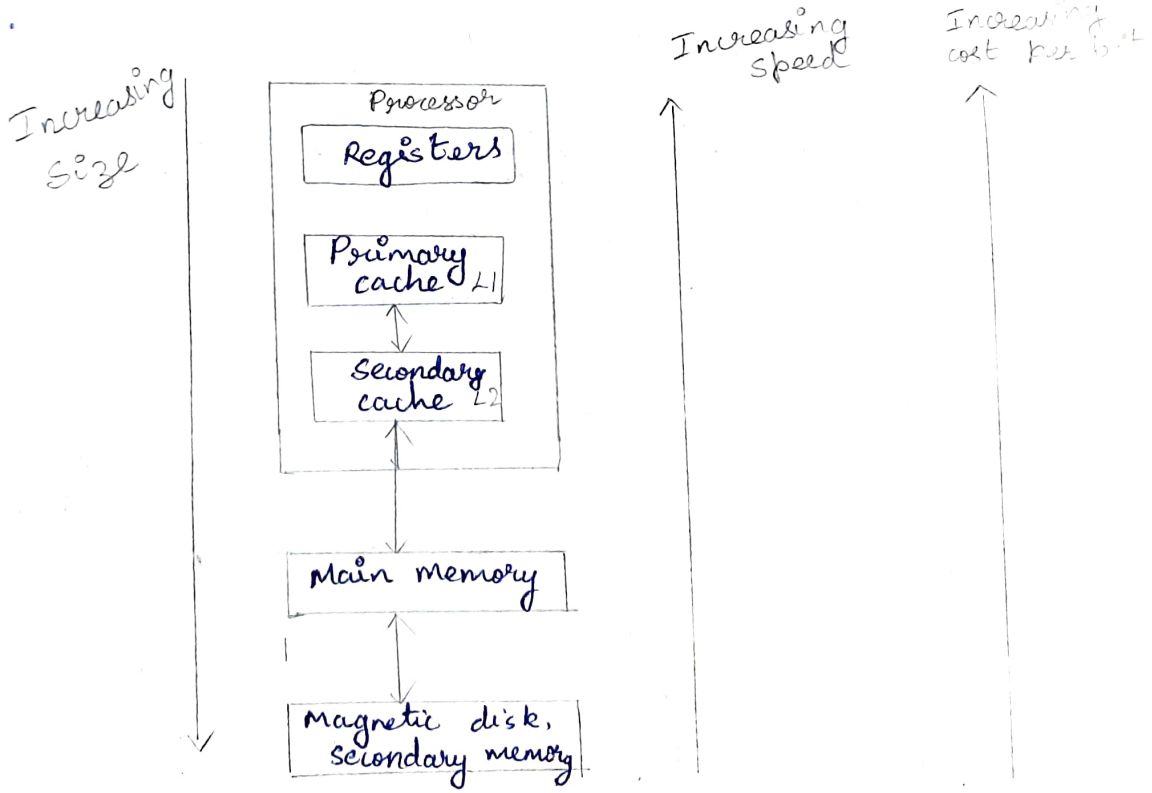
Ans:

```
LOAD        R2, #PACKED     Loading packed in R2
                            R2 points to PACKED
LoadByte    R3, (R2)        Stores Packed value to R3.
AND         R4, R3, #0xF    Performing and operation
                            (Last 4 bits)
Add         R4, #30         converts to BCD
StoreByte   R4, LOC         Storing it in LOC.
Clear       R4              clears R4
AND         R4, R3, #Fx0    And operation is done
                            and stored in R4
```

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LShiftR	R_4, #4	Shifts the first four bits to last
Add	R_4, R_4, #30	Converts to BCD
StoreByte	R_4, LOC+1	stores the result in LOC+1

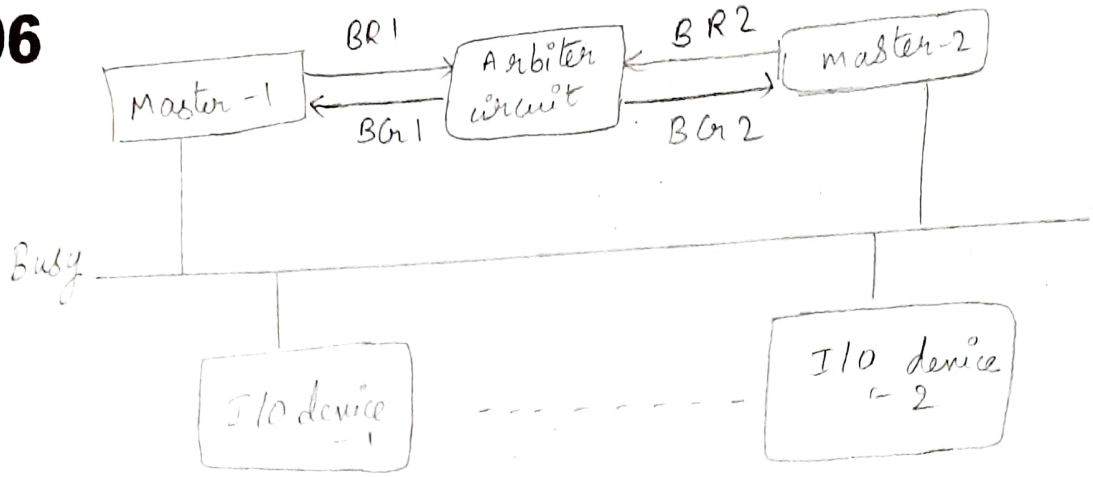
4.) With a neat diagram explain the concept of memory Hierarchy.



* Fastest acess is to data held in processor registers. Registers are at the top of the memory hierarchy
* Relatively small amount of memory that can be implemented on the processor chip. this is processor cache.
* Two levels of cache. Level 1 (L1) cache is on the processor chip. Level 2 (L2) cache is in between main memory and processor
* Next level is main memory, implemented as SIMMS. Much larger, but much slower than cache memory.
* Next level is magnetic disks. Huge amount of inexpensive storage.
* Speed of memory access is critical, the idea is to bring instructions and data that will be used in the near future as closer to the processor as possible.

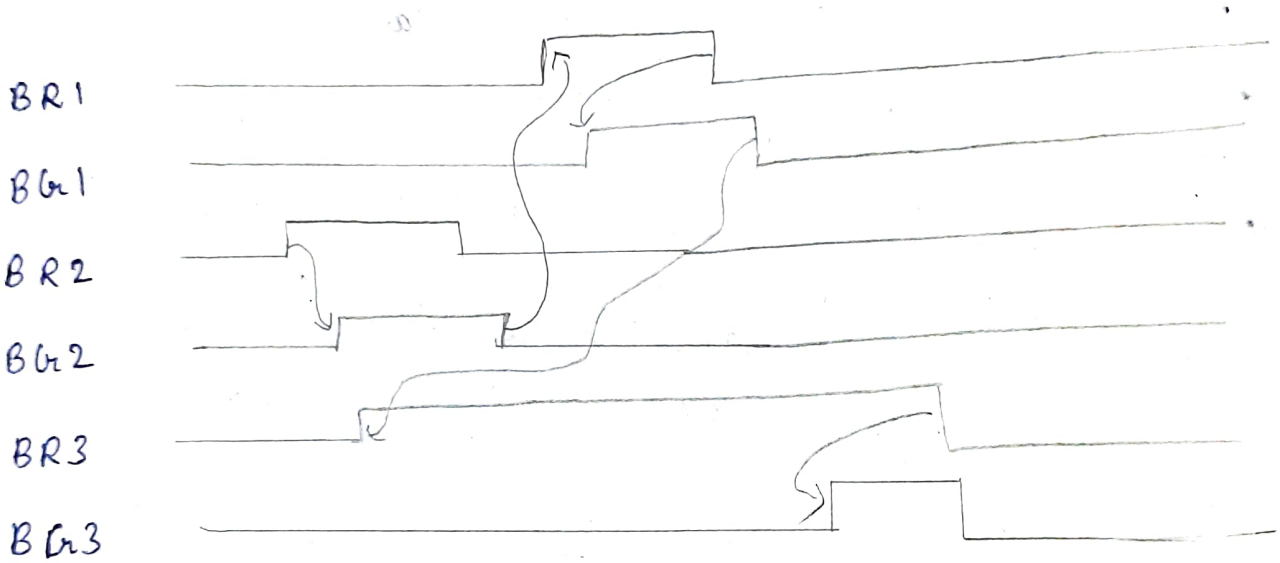
5.) In the bus arbiter protocol based on priority, the master that receives a bus grant maintains its request line in asserted state until it is ready to relinquish bus mastership. Assume that a common line called Busy is available, which is asserted by the master that is currently using the bus. The arbiter grants the bus only when the Busy is inactive. Once a master is receives a grants, it asserts Busy and drops its request, and in response the arbiter drops the grant. The master deactives Busy when it is finished using the bus. Draw a timing diagr for this mode of operation.

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Bus arbitration

Timing diagram:



Granting use of bus based on priority.

Here master 1 has the highest priority followed by other in numerical order.

6.) A computer system has a 1GB main memory. It also has a 4K-Byte cache organized as a 4 way set associative, with 4 blocks per set and 64 bytes per block. Calculate the number of bits in the tag, set index and byte offset fields

of the memory address format.

Ans: Main memory = 1GB = 2^{30}

∴ No. of bits in address line = 30.

Cache memory = 4K bytes = 2^{12}

1 block = 64 bytes i.e; 64 words = 2^6

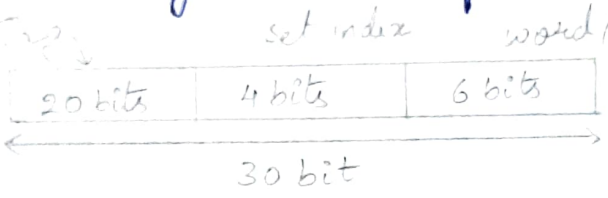
∴ word offset = 6 bits

No. of lines = $2^{12}/2^6 = 2^6$

No. of sets = $2^6/2^2$ (4-way associate) = 2^4

∴ Set bits = 4.

The memory address format is:



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7) Write a note on different cache write strategies:

Ans ① Cache hit:

The write operations is done in two ways.

ⓐ Write through protocol

ⓑ Write back protocol

ⓐ Write through protocol:

Here, the cache and main memory are updated simultaneously.

ⓑ Write-back protocol:

This technique is to update the cache location only and mark the cache location with an associated flag bit called dirty/modified bit.

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The word in memory will be updated later, when the marked block is removed from cache.

(ii) Write miss:

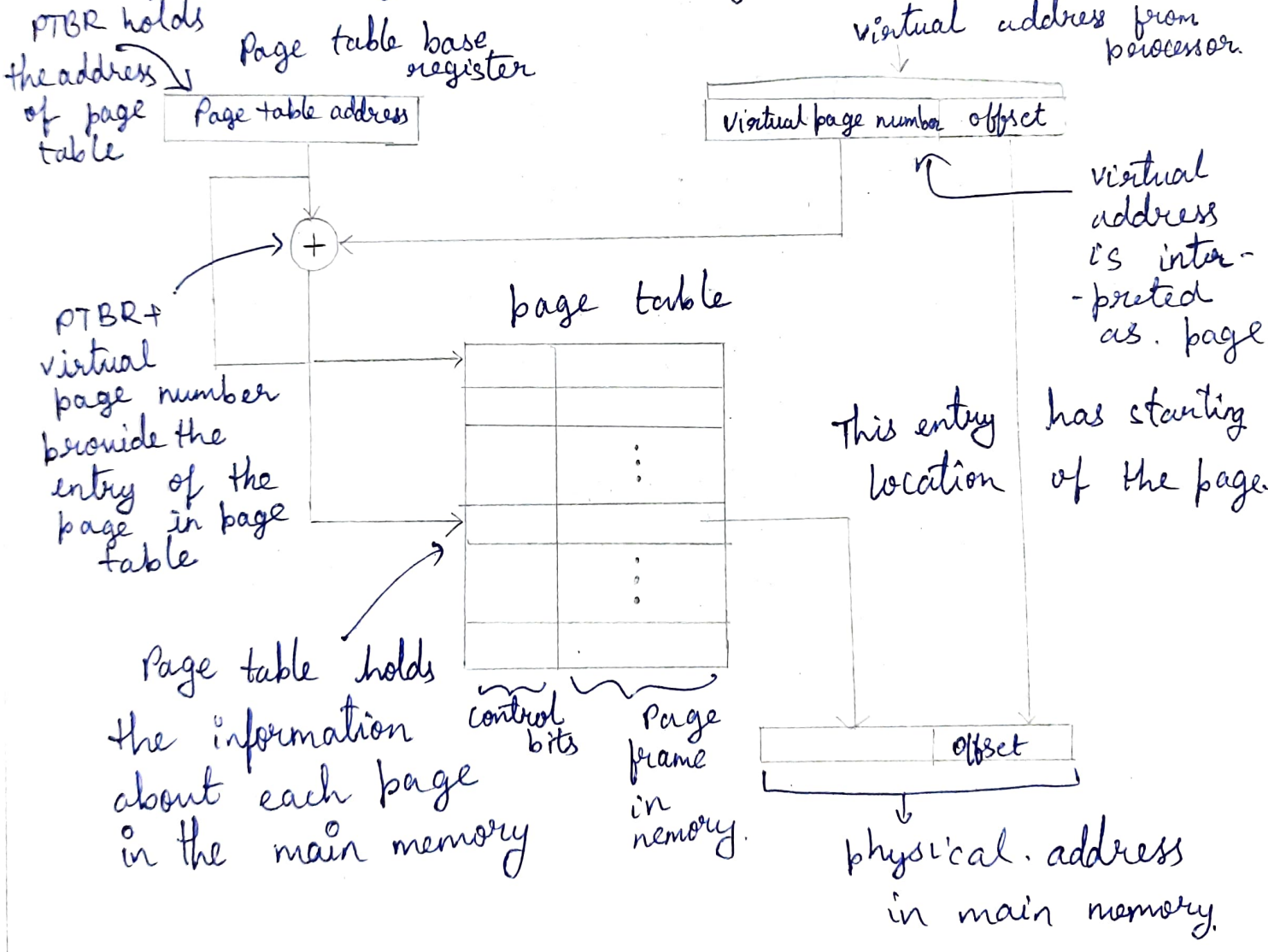
If required word does not exist, write miss occurs. If write through protocol is used, the main memory is updated automatically. But if write back protocol is used, block containing the addressed word is first brought into cache and then the desired word in the cache is overwritten with new information.

8.) Describe address translation in virtual memory with a neat diagram.

Ans:

* Each virtual or logical address generated by a processor is interpreted as a virtual page number (high-order bits) plus an offset (low-order bits) that specifies the location of a particular byte information within that page.
* Information about the main memory location of each page is kept in the page table.
* Area of main memory that can hold a page is called page frame.
* Starting address of the page table is kept in a page table base register.
* Virtual page number generated by the processor is added to the contents of the page table base register.

* The contents of this location in the page table give the starting address of the page if the page is currently in the main memory.
* Page table entry for a page also includes some control bits which describe the status of the page while it is in the main memory.
* One bit indicates the validity of the page.
* One bit indicates whether the page has been modified during its residency in the main memory.



9.) What is the difference between:

(i) Subroutine and interrupt service routine.

Interrupt service routine	Subroutine
* The interrupt is usually initiated by an internal or a external signal microprocessor rather than the execution of instruction.	* The subroutine is invoked by instructions which perform the specific tasks, and also reduces the size of program.
* After storing the current status of program with stack, the ISR is executed. ISR performs different types of tasks depending on the device which interrupted or instructions written by its programmer.	* Subroutine executes only the instructions given by the programmer
* The address of ISR is determined by the hardware	* Address of the subroutine is written inside the main program
* Call can be coming unexpectedly	* Call is gotten completely expectedly.

(ii) Memory mapped I/O and direct I/O

Memory mapped I/O	Direct I/O
* Main memory & I/O device share same address space.	* Seperate I/O device share address
* Machine instructions that access memory can be used to transfered data to and from I/O devices RISC: Load, Store.	* seperate instruc^n are used to perform I/O operations Ex: In, Out, ins, ~~outs~~
* Part of system memory is used for I/O device mapping.	* Complete system memory is used for storing data and instruction.
* More complex address decoding hardware are req.	* Simple address decoding hardware

(iii) Synchronous and asynchronous DRAM:

Synchronous DRAM	Asynchronous DRAM
* A DRAM type that uses an externally supplied clock signal to coordinate the operation of its external pin interfaces.	* An older type of DRAM used in earlier personal computers.
* System clock coordinates the memory access.	* No system clock is used for memory access.
* Provides high performance	* Provides low performance.
* Refresher circuits are used	* Refresher circuits are not used
* Manufacturing synchronous DRAM is high	* Manufacturing is relatively rare
* Used in high speed computers.	* Used for low speed.

(iv) Interrupt and Exception

Interrupt	Exception
* These are hardware interrupts	* these are software interrupts.
* Occurances usually disrupt other hardware interrupts	* Not true in case of exception.
* Asynchronous external requests	* Synchronous internal requests
* Can occur anywhere in the program	* Occur when there is an abnormal event in the program.
* Should not interfere with normal running of a program	* Abnormal events and often result in termination of a program.

(v) SRAM and DRAM

SRAM	DRAM
* Transistors are used to store data	* Capacitors are used to store data
* No refreshing is required	* Refreshing is required periodically.
* Faster when compared to DRAM	* It is slower. compared to SRAM
* They are expensive	* They are cheaper
* SRAMs are low density devices	* DRAMs are high density devices
* Used in cache memories	* Used in main memory.